



BIRZEIT UNIVERSITY

Faculty of Engineering & Technology
Department of Electrical & Computer Engineering
ENCS4370-COMPUTER ARCHITECTURE

Project #1
Test Cases

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Section:

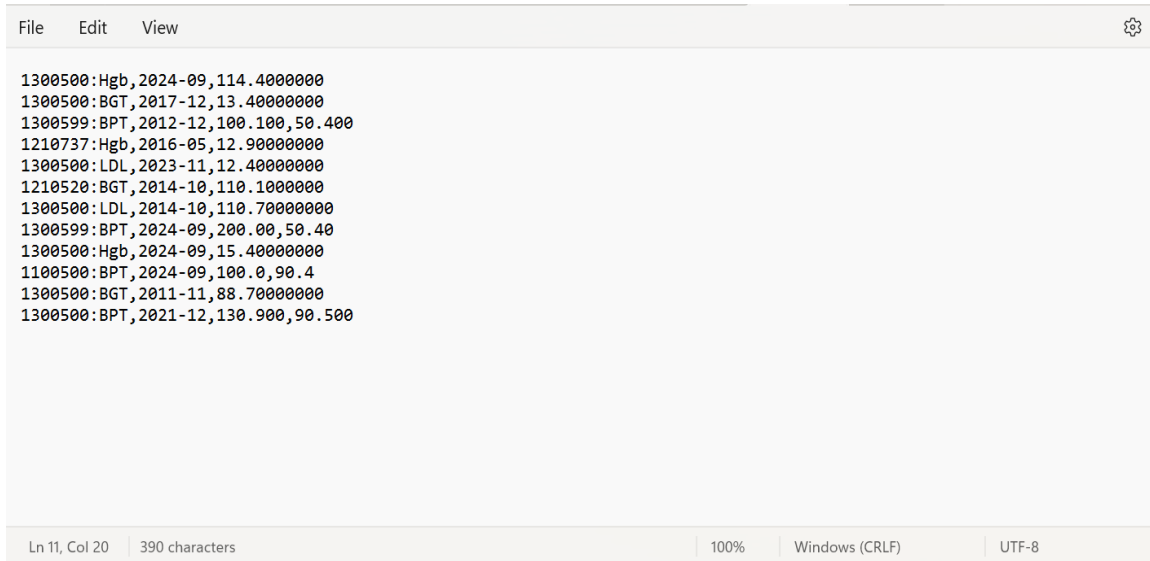
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Date:

28/4/2024

Table of Figures:

Figure1 :medical test inside our file.....	2
Figure 2:Search for a test by patient ID (Retrieve all patient tests).....	2
Figure 3:Add a new medical test with check the validity of the input data.....	3
Figure 4:Search for a test by patient ID(Retrieve all up normal patient tests)	3
Figure 5:Average test value 1	4
Figure 6:Update an existing test result.....	4
Figure 7:Average test value 2	5
Figure 8>Delete a test	6
Figure 9:search for the test we delete	7
Figure 10:Search for a test by patient ID(Retrieve all patient tests in a given specific period)	8
Figure 11:Searching for unnoraml tests based on the medical input test	9
Figure 12:Exit the program.....	9

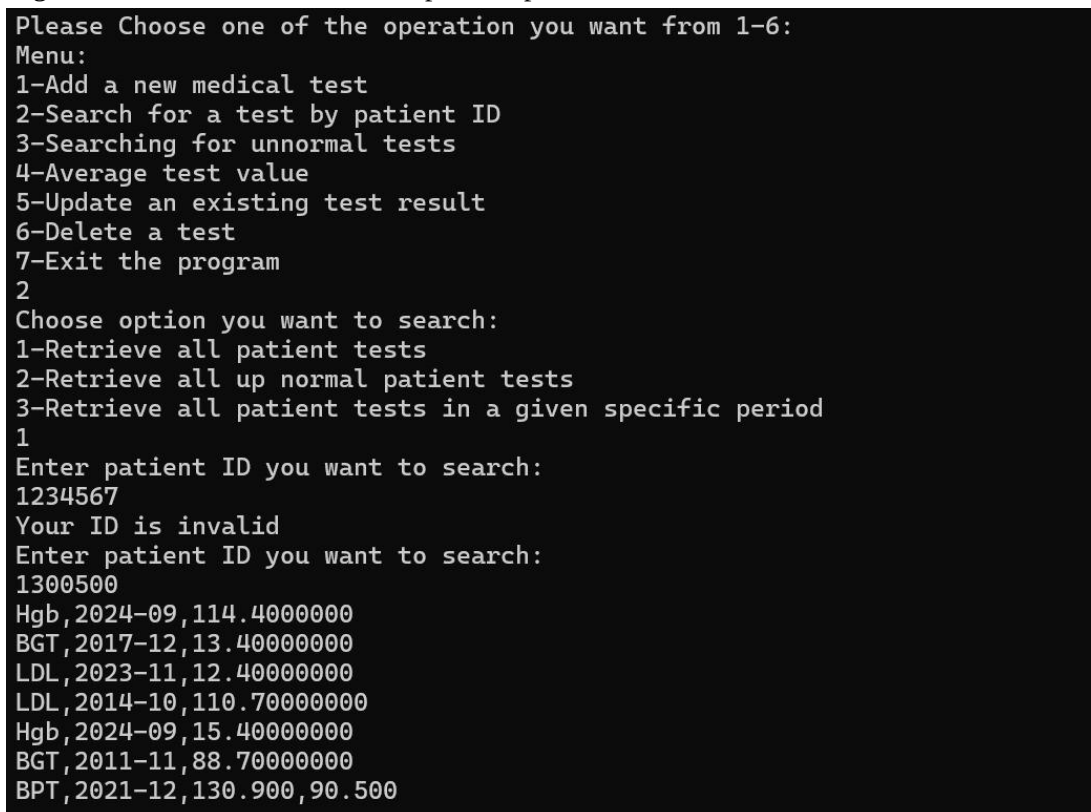


The screenshot shows a text editor window with a menu bar (File, Edit, View) and a settings icon. The main area contains a list of medical test records. The status bar at the bottom indicates 'Ln 11, Col 20', '390 characters', '100%', 'Windows (CRLF)', and 'UTF-8'.

```
1300500:Hgb,2024-09,114.4000000
1300500:BGT,2017-12,13.40000000
1300599:BPT,2012-12,100.100,50.400
1210737:Hgb,2016-05,12.90000000
1300500:LDL,2023-11,12.40000000
1210520:BGT,2014-10,110.1000000
1300500:LDL,2014-10,110.70000000
1300599:BPT,2024-09,200.00,50.40
1300500:Hgb,2024-09,15.40000000
1100500:BPT,2024-09,100.0,90.4
1300500:BGT,2011-11,88.70000000
1300500:BPT,2021-12,130.900,90.500
```

Figure 1 :medical test inside our file

First we search about all patient test by enter invalid id which is not found in medical test so let the user enter it again , then when enter valid id →print all patient test with this id



The screenshot shows a terminal window with a menu-driven program. The user selects option 2 to search for a test by patient ID. They enter an invalid ID (1234567) and are prompted to enter a valid one (1300500). The program then displays all medical test records for that patient ID.

```
Please Choose one of the operation you want from 1-6:
Menu:
1-Add a new medical test
2-Search for a test by patient ID
3-Searching for unnormal tests
4-Average test value
5-Update an existing test result
6-Delete a test
7-Exit the program
2
Choose option you want to search:
1-Retrieve all patient tests
2-Retrieve all up normal patient tests
3-Retrieve all patient tests in a given specific period
1
Enter patient ID you want to search:
1234567
Your ID is invalid
Enter patient ID you want to search:
1300500
Hgb,2024-09,114.4000000
BGT,2017-12,13.40000000
LDL,2023-11,12.40000000
LDL,2014-10,110.70000000
Hgb,2024-09,15.40000000
BGT,2011-11,88.70000000
BPT,2021-12,130.900,90.500
```

Figure 2:Search for a test by patient ID (Retrieve all patient tests)

Then we add a new medical test but with invalid input so we print a msg to user to reenter valid medical test to add

```

Please Choose one of the operation you want from 1-6:
Menu:
1-Add a new medical test
2-Search for a test by patient ID
3-Searching for unnormal tests
4-Average test value
5-Update an existing test result
6-Delete a test
7-Exit the program
1
Enter a single medical test which should be in the following representation where Patient Id is 7 digit without char
Test Name is 3 char without numbers
year in range from 1950 to 2050 without char
month with range from 01 to 12 without char
and Result in float representation without char:
Patient ID: TestName, TestDate, result
123456h:BG6,2o23-o9,12.o6
The Patient ID is not valid , please Enter ID with 7 digit(int)
The test Name is not valid ,please Enter Test Name with 3 character
The year you enter is not valid,please enter year in range  from 1950 to 2050
The month is not valid,please enter month in range from 01 to 12
The result you enter is invalid please enter it without char
Enter a single medical test which should be in the following representation where Patient Id is 7 digit without char
Test Name is 3 char without numbers
year in range from 1950 to 2050 without char
month with range from 01 to 12 without char
and Result in float representation without char:
Patient ID: TestName, TestDate, result
1210737:Hgb,2024-04,100.879
Your Test Has been added.

```

Figure 3:Add a new medical test with check the validity of the input data

Search for a medical test we add by search about unnormal test for test we add

```

Please Choose one of the operation you want from 1-6:
Menu:
1-Add a new medical test
2-Search for a test by patient ID
3-Searching for unnormal tests
4-Average test value
5-Update an existing test result
6-Delete a test
7-Exit the program
2
Choose option you want to search:
1-Retrieve all patient tests
2-Retrieve all up normal patient tests
3-Retrieve all patient tests in a given specific period
2
Enter patient ID you want to search:
1210737
Hgb,2016-05,12.9
Hgb,2024-04,100.879

```

Figure 4:Search for a test by patient ID(Retrieve all up normal patient tests)

Then Print the avg for all test in medical test

```

Please Choose one of the operation you want from 1-6:
Menu:
1-Add a new medical test
2-Search for a test by patient ID
3-Searching for unnormal tests
4-Average test value
5-Update an existing test result
6-Delete a test
7-Exit the program
4
Hgb:60.89475
LDL:61.55
BGT:70.73333
BPT:132.75,70.425

```

Figure 5:Average test value 1

Then we want to update and enter test that not found so we ask the user to reenter test to update the result

```

Please Choose one of the operation you want from 1-6:
Menu:
1-Add a new medical test
2-Search for a test by patient ID
3-Searching for unnormal tests
4-Average test value
5-Update an existing test result
6-Delete a test
7-Exit the program
5
1300500:Hgb,2024-09,114.40000000
1300500:BGT,2017-12,13.40000000
1300599:BPT,2012-12,100.100,50.400
1210737:Hgb,2016-05,12.90000000
1300500:LDL,2023-11,12.40000000
1210520:BGT,2014-10,110.10000000
1300500:LDL,2014-10,110.70000000
1300599:BPT,2024-09,200.00,50.40
1300500:Hgb,2024-09,15.40000000
1100500:BPT,2024-09,100.0,90.4
1300500:BGT,2011-11,88.70000000
1300500:BPT,2021-12,130.900,90.500
1210737:Hgb,2024-04,100.879
Enter the Whole Test you want to Update
1300590:BPT,2012-12,100.3,50.4
The Test you enter is not found
1300500:Hgb,2024-09,114.40000000
1300500:BGT,2017-12,13.40000000
1300599:BPT,2012-12,100.100,50.400
1210737:Hgb,2016-05,12.90000000
1300500:LDL,2023-11,12.40000000
1210520:BGT,2014-10,110.10000000

```

```

1210737:Hgb,2016-05,12.90000000
1300500:LDL,2023-11,12.40000000
1210520:BGT,2014-10,110.10000000
1300500:LDL,2014-10,110.70000000
1300599:BPT,2024-09,200.00,50.40
1300500:Hgb,2024-09,15.40000000
1100500:BPT,2024-09,100.0,90.4
1300500:BGT,2011-11,88.70000000
1300500:BPT,2021-12,130.900,90.500
1210737:Hgb,2024-04,100.879
Enter the Whole Test you want to Update
1300590:BPT,2012-12,100.3,50.4
The Test you enter is not found
1300500:Hgb,2024-09,114.40000000
1300500:BGT,2017-12,13.40000000
1300599:BPT,2012-12,100.100,50.400
1210737:Hgb,2016-05,12.90000000
1300500:LDL,2023-11,12.40000000
1210520:BGT,2014-10,110.10000000
1300500:LDL,2014-10,110.70000000
1300599:BPT,2024-09,200.00,50.40
1300500:Hgb,2024-09,15.40000000
1100500:BPT,2024-09,100.0,90.4
1300500:BGT,2011-11,88.70000000
1300500:BPT,2021-12,130.900,90.500
1210737:Hgb,2024-04,100.879
Enter the Whole Test you want to Update
1300599:BPT,2012-12,100.100,50.400
Enter the New Result for test you choose one
One result if the test is Hgb or LDL or BGT
Two Result if the Test is BPT with Comma between them:
200.678,67.23
The Test you choose is Updated

```

Figure 6:Update an existing test result

Then we print all the avg again to ensure that the test we choose updated

```
Please Choose one of the operation you want from 1-6:
Menu:
1-Add a new medical test
2-Search for a test by patient ID
3-Searching for unnormal tests
4-Average test value
5-Update an existing test result
6-Delete a test
7-Exit the program
4
Hgb:60.89475
LDL:61.55
BGT:70.73333
BPT:157.8945,74.6325
```

Figure 7:Average test value 2

Then we delete a test by enter a test that not found so ask the user to reenter it then it was deleted

Please Choose one of the operation you want from 1-6:

Menu:

- 1-Add a new medical test
- 2-Search for a test by patient ID
- 3-Searching for unnormal tests
- 4-Average test value
- 5-Update an existing test result
- 6-Delete a test
- 7-Exit the program

6

1300500:Hgb,2024-09,114.4000000
1300500:BGT,2017-12,13.40000000
1300599:BPT,2012-12,200.678,67.230
1210737:Hgb,2016-05,12.90000000
1300500:LDL,2023-11,12.40000000
1210520:BGT,2014-10,110.1000000
1300500:LDL,2014-10,110.70000000
1300599:BPT,2024-09,200.00,50.40
1300500:Hgb,2024-09,15.40000000
1100500:BPT,2024-09,100.0,90.4
1300500:BGT,2011-11,88.70000000
1300500:BPT,2021-12,130.900,90.500
1210737:Hgb,2024-04,100.879

Enter the whole Test to Delete:

1234567:Hgb,2012-09,67.89

The Test you enter is not found

1300500:Hgb,2024-09,114.4000000
1300500:BGT,2017-12,13.40000000
1300599:BPT,2012-12,200.678,67.230
1210737:Hgb,2016-05,12.90000000
1300500:LDL,2023-11,12.40000000
1210520:BGT,2014-10,110.1000000
1300500:LDL,2014-10,110.70000000

6

1300500:Hgb,2024-09,114.4000000
1300500:BGT,2017-12,13.40000000
1300599:BPT,2012-12,200.678,67.230
1210737:Hgb,2016-05,12.90000000
1300500:LDL,2023-11,12.40000000
1210520:BGT,2014-10,110.1000000
1300500:LDL,2014-10,110.70000000
1300599:BPT,2024-09,200.00,50.40
1300500:Hgb,2024-09,15.40000000
1100500:BPT,2024-09,100.0,90.4
1300500:BGT,2011-11,88.70000000
1300500:BPT,2021-12,130.900,90.500
1210737:Hgb,2024-04,100.879

Enter the whole Test to Delete:

1234567:Hgb,2012-09,67.89

The Test you enter is not found

1300500:Hgb,2024-09,114.4000000
1300500:BGT,2017-12,13.40000000
1300599:BPT,2012-12,200.678,67.230
1210737:Hgb,2016-05,12.90000000
1300500:LDL,2023-11,12.40000000
1210520:BGT,2014-10,110.1000000
1300500:LDL,2014-10,110.70000000
1300599:BPT,2024-09,200.00,50.40
1300500:Hgb,2024-09,15.40000000
1100500:BPT,2024-09,100.0,90.4
1300500:BGT,2011-11,88.70000000
1300500:BPT,2021-12,130.900,90.500
1210737:Hgb,2024-04,100.879

Enter the whole Test to Delete:

1100500:BPT,2024-09,100.0,90.4

The Test you choose is Deleted

Figure 8:Delete a test

Then we search about the test we deleted by search for all patient test by there id , so print that it not found and search about another id

```
Please Choose one of the operation you want from 1-6:
Menu:
1-Add a new medical test
2-Search for a test by patient ID
3-Searching for unnormal tests
4-Average test value
5-Update an existing test result
6-Delete a test
7-Exit the program
2
Choose option you want to search:
1-Retrieve all patient tests
2-Retrieve all up normal patient tests
3-Retrieve all patient tests in a given specific period
1
Enter patient ID you want to search:
1100500
Your ID is invalid
Enter patient ID you want to search:
1210737
Hgb,2016-05,12.900000000
Hgb,2024-04,100.879
```

Figure 9:search for the test we delete

Then we search about all patient tests in a given specific period by there id

```
Please Choose one of the operation you want from 1-6:
Menu:
1-Add a new medical test
2-Search for a test by patient ID
3-Searching for unnormal tests
4-Average test value
5-Update an existing test result
6-Delete a test
7-Exit the program
2
Choose option you want to search:
1-Retrieve all patient tests
2-Retrieve all up normal patient tests
3-Retrieve all patient tests in a given specific period
3
Enter patient ID you want to search:
1234567
Enter a year the first year:
2010
Enter a year the second year:
2017
Your ID is invalid
Enter patient ID you want to search:
1300500
Enter a year the first year:
2010
Enter a year the second year:
2017
BGT,2017-12,13.40000000
LDL,2014-10,110.70000000
BGT,2011-11,88.70000000
```

Figure 10:Search for a test by patient ID(Retrieve all patient tests in a given specific period)

Search for unormal test(for test we updated)

```
Please Choose one of the operation you want from 1-6:
Menu:
1-Add a new medical test
2-Search for a test by patient ID
3-Searching for unnormal tests
4-Average test value
5-Update an existing test result
6-Delete a test
7-Exit the program
3
Enter Name of Test you want to search
BPT
1300599:BPT,2012-12,200.678,67.23
1300599:BPT,2024-09,200.0,50.4
1300500:BPT,2021-12,130.9,90.5
```

Figure 11:Searching for unnormal tests based on the medical input test

Then we exit the program ☺

```
Please Choose one of the operation you want from 1-6:
Menu:
1-Add a new medical test
2-Search for a test by patient ID
3-Searching for unnormal tests
4-Average test value
5-Update an existing test result
6-Delete a test
7-Exit the program
7

C:\Users\DELL\Desktop\Arc\project1>
```

Figure 12:Exit the program